

Process Requirement Specification - NaCl Evaporation Plant

Document status: Released Rev A. Later approved change records may override individual values or routing statements in this document.

1. Purpose and process intent

The laboratory plant shall concentrate a water/sodium-chloride batch using staged charging, mixing, buffering, evaporation, condensation, cooling and return pumping. B1 is initially charged with water and B2 with concentrated brine. B3 prepares the batch, B4 buffers it, B5 evaporates water, K1 condenses the vapor, B6 cools condensate and B7 cools concentrate.

2. High-level process sequence

Phase	Required action	Completion condition
Charge water	Open V8, B1 to B3	LIS-301 $\geq$ 0.13 m
Charge brine / mix	Open V9, B2 to B3	QI-302 reaches recipe target
Buffer transfer	Open V11, B3 to B4	LIS-301 $<$ 0.01 m
Evaporator charge	When B5 is idle, open V12, B4 to B5	LIS-501 reaches batch level
Evaporation	Enable B5 heater	QIS-502 reaches evaporation target
Condensate branch	Cool B6, then return condensate	Cooling complete and B6 returned
Concentrate branch	Transfer B5 to B7, cool B7, return concentrate	Cooling complete and B7 returned
Cycle reset	Wait for both branches	Both complete and time $>$ 2500 s

3. Equipment and control requirements

The central controller shall command discrete on/off valves. The P&ID; includes both controller-operated and manual/local valves. Automated valves shall fail closed. Pumps P1 and P2 return cooled process streams to the charging-vessel header system.

The intended physical simulation medium is a water-NaCl mixture. A StandardWater configuration may be used as an integration baseline to verify topology and sequence behavior before enabling composition-dependent property functions.

4. Modeling constraints

Tank connections can pass above or below the liquid level. The preferred numerical treatment is a regularized asymmetric loss-factor formulation with hysteresis around the port elevation. Small junction volumes shall be used where multiple isolating pressure-drop components could otherwise leave undefined pressure, enthalpy or composition states.

All non-horizontal pipes shall include static head in their pressure-drop relationship. The evaporator/condenser may be represented as a combined simulation component provided the physical condenser K1 remains identifiable in the system architecture.

5. Verification

The acceptance simulation shall demonstrate a full charge-mix-buffer-evaporate-cool-return cycle, including the controller parallel split and join after evaporation. It shall continue beyond 2500 s to demonstrate cycle reset.